

The software for NA62 is split into three main C++ packages: NA62MC, NA62Reconstruction and NA62Analysis. NA62MC is a GEANT4 based Monte Carlo simulation package that simulates the NA62 experiment including the detector geometry and the full simulation of kaon decays, used as a comparison to the data collected by the experiment. NA62Reconstruction is a package that is used to reconstruct either the Monte Carlo data or collected data from the experiment for all detectors. NA62Analysis is used by the physicists to analyse the data produced from the reconstruction to produce all the physics studies that are conducted at NA62.

The NA62 software was designed for use only with the NA62 experiment resulting in minimal configurability and flexibility, with the only option for users to change the software being to enable/disable sub-detectors. With the experiment foreseen to run up to CERN's Long shutdown 3 (LS3), an upgraded software package was needed to allow for the development of new detectors for the future kaon experiments. This software had to be flexible and configurable to allow for the simultaneous use of the software for NA62 data taking and physics analysis, as well as the development of new experiments.

0.1 FlexibleMC

For the implementation of a flexible framework all detector classes were refactored to inherit from a common class which contained the basic functions and variables for all detectors allowing for all classes to be parsed using the same machinery. The input to the simulation was then changed to allow for the configuration of all the detectors to be read in from a configuration file which contained any variables required for individual detectors and the X,Y,Z and rotation of the detectors. An

example entry in the configuration file is shown in Figure 1 where a name was given for the station "STRAW0", along with chamber ID, chmaber type which defines the size of the straws and the chamber geometry which defines if the beam hole is symetric or not. The variables for each subdetector are defined in Table

Name	Arguments	X	Y	Z	θ_x	θ_y	θ_z
Spectrometer	("STRAW0", 0, 0, 0)	0	0	183705.9	0	0	0

Figure 1: Example configuration for the STRAW0

Detector Name	Variables
STRAW	Chamber ID: 0,1,2,3
	Chamber Type: 0(10mm), 1(5mm)
	Chamber Mode: 0(default), 1(symmetric)
Spectrometer Magnet	Magnetic field intensity
Vessel	Length
	InnerRadius
	OuterRadius
	Type("BlueTube", "BlueTubeNoRails", "GreyTube")

Table 1

When built inside the simulation, each detector was now contained within its own mother volumen called the "Responsibilty Region" which was just slighly larger than the detector volume itself. This was a simple task for most detectors apart from the STRAWs and LAV which were made up of multiple stations that were not continuos, resulting in mutliple responsibility regions for each station.

0.2 SlimMC

Bibliography